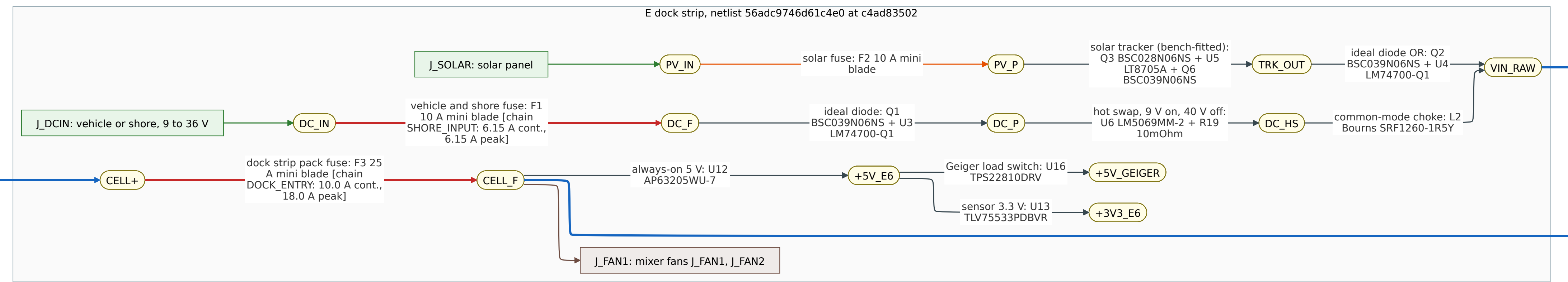


Rounded nodes are nets as the netlists name them (each board's own names; board B's VBAT_RTC is its CR2032 net, VBAT until W3B-R1). Red edges are stages of pcb_energy_chain.yaml with its declared continuous and peak current; orange, a fuse or polyfuse (as the netlist gives the part, never by its designator); grey, any other series stage: a converter, load switch, eFuse, ideal diode, hot swap, current shunt, ferrite bead or choke; blue, a lead between boards. Which parts make each stage (the part names on each edge) is typed in tools/power_tree.py, not found by it, every build checks that typing against the netlist: the path runs only through the named parts; the FETs a named controller IC drives and the inductors on their switching nodes, each named part is required, a controller counts only through the FET gates it drives on the path, an enable named in brackets is met by a named IC, and 2927 wrong entries (swapped controllers, shunts and enables among them) are refused. The attribution check is not complete: a part bridging a stage's two nets (a monitor, a feedback resistor, a load) passes in place of the stage's own part (10 of 2715 substitutions tried at this build), so the parts named were also read by hand. It does not check that a stage works; power-tree.md has the method and its limits. NOT SHOWN: currents drawn, losses and heat (POWER-THERMAL.md), which rail is on in which power state, grounds and returns, the CMS modules' internal rails, clamps and bulk capacitors. The chain file does not yet carry PWR-F12's F2 stage (its pack follows D-06 since @b1d4); see power-tree.md. Nothing here is built, powered or measured.



E5 dock block: no schematic, its board names the nets

